

Smart Factory Assistant: AI-Based Predictive Maintenance and Job Scheduling System for Smart Manufacturing

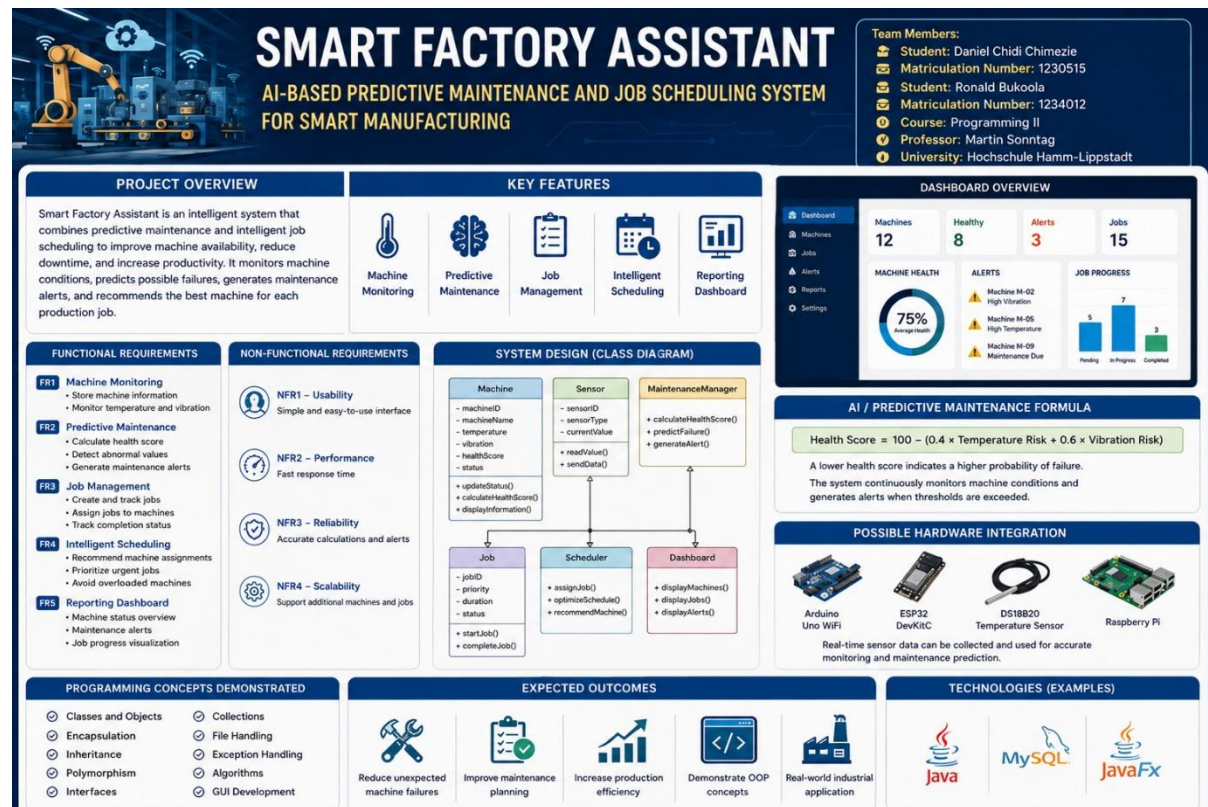
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1. Project Overview

The proposed project is a Smart Factory Assistant that combines predictive maintenance and intelligent job scheduling for manufacturing environments. The system will monitor machine conditions, analyze operational data, predict potential failures, and optimize the assignment of production jobs to available machines.

The goal is to improve machine availability, reduce downtime, increase productivity, and demonstrate the practical application of object-oriented programming concepts learned in Programming II.

2. Objectives

The system aims to:

- Monitor machine operating conditions.
- Predict possible machine failures before they occur.

- Generate maintenance alerts automatically.
- Manage production jobs efficiently.
- Recommend optimal machine assignments.
- Improve overall factory productivity.

3. Functional Requirements

FR1 – Machine Monitoring

The system shall:

- Store machine information.
- Monitor machine status.
- Record temperature readings.
- Record vibration measurements.
- Display machine health information.

FR2 – Predictive Maintenance

The system shall:

- Calculate machine health scores.
- Detect abnormal operating conditions.
- Generate maintenance warnings.
- Recommend maintenance actions.

FR3 – Job Management

The system shall:

- Create production jobs.
- Track job status.
- Assign jobs to machines.
- Record job completion.

FR4 – Intelligent Scheduling

The system shall:

- Recommend suitable machines for jobs.
- Prioritize urgent tasks.
- Avoid assigning jobs to unhealthy machines.
- Optimize machine utilization.

FR5 – Reporting Dashboard

The system shall:

- Display machine information.
- Display maintenance alerts.
- Display job progress.
- Generate summary reports.

4. Non-Functional Requirements

NFR1 – Usability

The user interface shall be simple and easy to understand.

NFR2 – Performance

The system shall respond quickly to user interactions.

NFR3 – Reliability

The system shall provide consistent and accurate results.

NFR4 – Scalability

The system shall support the addition of more machines and jobs in the future.

5. Object-Oriented Programming Concepts

The project demonstrates the following Programming II concepts:

- Classes and Objects
- Encapsulation
- Inheritance
- Polymorphism
- Interfaces
- Collections
- Exception Handling
- File Handling
- Algorithms
- Graphical User Interface Development

6. Proposed System Components

Machine Class

Attributes:

- machineID
- machineName
- temperature
- vibration
- healthScore
- status

Methods:

- updateStatus()
- calculateHealthScore()
- displayInformation()

Sensor Class

Attributes:

- sensorID
- sensorType
- currentValue

Methods:

- readValue()
- sendData()

MaintenanceManager Class

Methods:

- calculateHealthScore()
- predictFailure()
- generateAlert()

Job Class

Attributes:

- jobID
- priority
- duration
- status

Methods:

- startJob()
- completeJob()

Scheduler Class

Methods:

- assignJob()
- optimizeSchedule()
- recommendMachine()

Dashboard Class

Methods:

- displayMachines()
- displayJobs()
- displayAlerts()

7. Predictive Maintenance Algorithm

The machine health score can be calculated using:

$$\text{Health Score} = 100 - (0.4 \times \text{Temperature Risk} + 0.6 \times \text{Vibration Risk})$$

A lower health score indicates a higher likelihood of machine failure.

The system will continuously monitor machine conditions and generate alerts whenever predefined thresholds are exceeded.

8. Future Hardware Integration

The software can later be integrated with hardware components such as:

- Arduino Uno WiFi
- ESP32 DevKitC
- Raspberry Pi
- DS18B20 Temperature Sensor

This would allow real-time monitoring and predictive maintenance demonstrations using actual sensor data.

9. Expected Results

The Smart Factory Assistant is expected to:

- Reduce unexpected machine failures.
- Improve maintenance planning.

- Increase production efficiency.
- Demonstrate advanced object-oriented programming concepts.
- Provide a realistic Industry 4.0 application suitable for Programming II.

10. Conclusion

The Smart Factory Assistant combines predictive maintenance and intelligent job scheduling into a single software solution. The project demonstrates important Programming II concepts while addressing real-world manufacturing challenges. It provides both academic value and practical industrial relevance, making it a suitable bonus project for Programming II.